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AdHImEx: Advection that is Accurate, Efficient, and Stable for Long Time Steps

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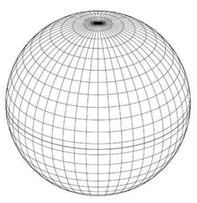
Next-generation dynamical cores need to take long time steps for efficiency. However, explicit Eulerian advection schemes have a strict time step restriction for stability. Adaptively implicit-explicit (AdImEx) schemes offer a solution, but have only been considered with first-order accuracy for large Courant numbers[1,2,3]. Here, we present AdHImEx: a new AdImEx scheme that increases the accuracy for large Courant numbers.

When Do Instabilities Arise?

$$\frac{\partial \Psi}{\partial t} + \nabla \cdot \left(\vec{u}\Psi\right) = 0$$

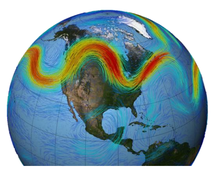
Numerical advection schemes calculate the tracer field Ψ over time t advected by the wind $\vec{u}$. The Courant number C represents how many grid boxes the advected property can travel in one time step. Explicit Eulerian schemes are typically unstable for $C > 1$. This can occur where there is grid convergence, strong winds, or convective updraughts.

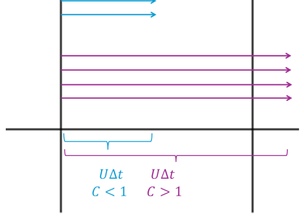
$$C = \frac{1}{2}\frac{\Delta t}{V}\sum_f |\vec{S}_f \cdot \vec{u}_f|$$



At the last grid point before the poles, $\Delta x = 10$ km at the midlatitudes is reduced to ~50 m with a lat-lon grid. With $u = 5$ m/s, Δt~10 s for $C = 1$.

u_{jet}~100 m/s with $\Delta x = 10$ km at the midlatitudes would require Δt~100 s for $C = 1$.





Adaptively Implicit-Explicit Time Stepping

Explicit schemes calculate the field at time step $n+1$ using only the field up until time level n, whereas implicit schemes also use the field at timestep $n+1$ to do so. Implicit schemes can be unconditionally stable, but inefficient. An **AdImEx** scheme treats advection explicitly where this is stable and (partially) implicitly otherwise, i.e.,

$$\text{Explicit: } \Psi^{(n+1)} = \Psi^{(n)} - \Delta t \hat{F}(\Psi^{(n)}, \Psi^{(n-1)}, \dots),$$
$$\text{Implicit: } \Psi^{(n+1)} = \Psi^{(n)} - \Delta t \tilde{F}(\Psi^{(n+1)}, \Psi^{(n)}, \Psi^{(n-1)}, \dots),$$
$$\text{AdImEx: } \Psi^{(n+1)} = \Psi^{(n)} - \Delta t(1-\theta)\hat{F}(\Psi^{(n)}, \Psi^{(n-1)}, \dots) - \Delta t\theta\tilde{F}(\Psi^{(n+1)}, \Psi^{(n)}, \Psi^{(n-1)}, \dots)$$

for some functions $\hat{F}$ and $\tilde{F}$ and implicitness θ, and can keep the model **stable for high C** by increasing θ from 0 toward 1, while remaining efficient for low C.

Previous work showed the potential of AdImEx schemes, but was only first-order accurate for high C[1,2,3] and can require multiple matrix solutions[2].

AdHImEx: A New AdImEx Scheme

The new Adaptively High-order Implicit-Explicit (AdHImEx) scheme uses Runge-Kutta multi-stage time stepping:

$$\Psi^s = \Psi^{(n)} - \Delta t \sum_{r=1}^{s-1} F_s\left(a_{sr}, \vec{u}^r, \Psi^r\right) - \Delta t F_s\left(a_{ss}, \vec{u}^s, \Psi^s\right), \qquad s = 1, ..., \sigma$$

$$\Psi^{(n+1)} = \Psi^{(n)} - \Delta t \sum_{s=1}^{\sigma} F\left(b_s, \vec{u}^s, \Psi^s\right),$$

c_1	a_{11}	0	$\cdots$	0
c_2	a_{21}	a_{22}	$\cdots$	0
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
c_σ	$a_{\sigma 1}$	$a_{\sigma 2}$	$\cdots$	$a_{\sigma\sigma}$
	b_1	b_2	$\cdots$	b_σ

where a, b are coefficients given in the Butcher tableau on the right, σ the total number of substages used, and s the specific substage.

AdHImEx, inspired by the UJ3(1+e,3,2) scheme[4], is

- Stable for $C \leq 100$++ with $\theta = 1 - \frac{1}{1+0.7\ \max(C-1.4,0)}$,
- Third-order accurate in time for $C \leq 1.4$,
- Second-order accurate in time for $C > 1.4$,
- Fifth-order accurate in space,
- Mass-conserving,
- Constancy-preserving, and has
- Only one implicit solve.

0	0	0	0	0	0
$\frac{1}{2}\theta$	$\frac{1}{2}\theta$	0	0	0	0
$1-\frac{1}{2}\theta$	$\frac{1}{2}\theta$	$1-\theta$	0	0	0
$\frac{1}{2}$	$\frac{1}{2}\theta$	$\frac{1}{4}(1-\theta)$	$\frac{1}{4}(1-\theta)$	0	0
1	$\frac{1}{2}\theta$	$\frac{1}{6}(1-\theta)$	$\frac{1}{6}(1-\theta)$	$\frac{2}{3}(1-\theta)$	$\frac{1}{2}\theta$
	$\frac{1}{2}\theta$	$\frac{1}{6}(1-\theta)$	$\frac{1}{6}(1-\theta)$	$\frac{2}{3}(1-\theta)$	$\frac{1}{2}\theta$

AdHImEx preserves constant fields with a stage-dependent treatment of the divergence:

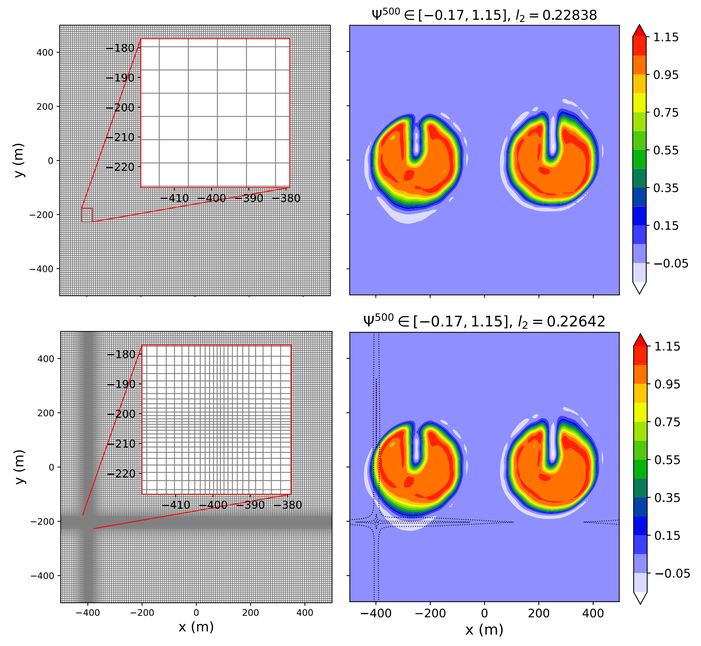
$$F_s\left(a_{sr}, \vec{u}, \Psi^r\right) = \begin{cases} \nabla \cdot \left(a_{sr}\vec{u}\Psi^r\right) & \text{for } s = 1, 4, 5, \\ a_{sr}\nabla \cdot \left(\vec{u}\Psi^r\right) & \text{for } s = 2, 3, \end{cases} \qquad F\left(b_s, \vec{u}, \Psi^s\right) = \nabla \cdot \left(b_s\vec{u}\Psi^s\right).$$

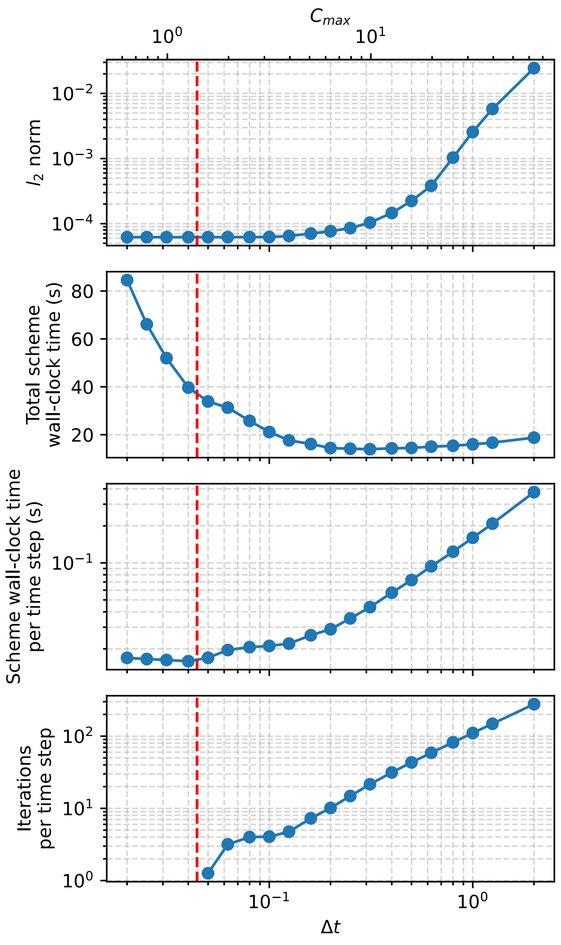
Deformational Flow

Nondivergent flow[5] is used to show the skill of the AdHImEx solution, with solutions on two grids. The upper row uses a uniform grid, such that $C_{max} = 0.8$ and the time stepping is explicit. The bottom row uses a nonuniform grid (with the same coarse resolution as the uniform grid), such that $C_{max} = 6.3$. This needs AdHImEx time stepping.

The $n_t = 500$ results ($\Delta t = 0.2$ s) show no visible deterioration in the AdHImEx solution compared to the explicit reference. The l_2 error norms virtually stay the same.

This excellent agreement confirms that AdHImEx remains accurate and reliable even with high C present, a significant challenge for many time stepping schemes.





Efficiency

Nondivergent flow[5], with the nonuniform grid but now with a double sine wave, is used to assess efficiency. Can it be more efficient to use AdHImEx instead of explicit time stepping? Yes.

The test is again run until T=100, but now for various Δt. The matrix solver is GMRES(m), restarting every four iterations.

As Δt increases, the wall-clock time per time step increases, but this is more than balanced out by the smaller number of time steps taken.

NWP test

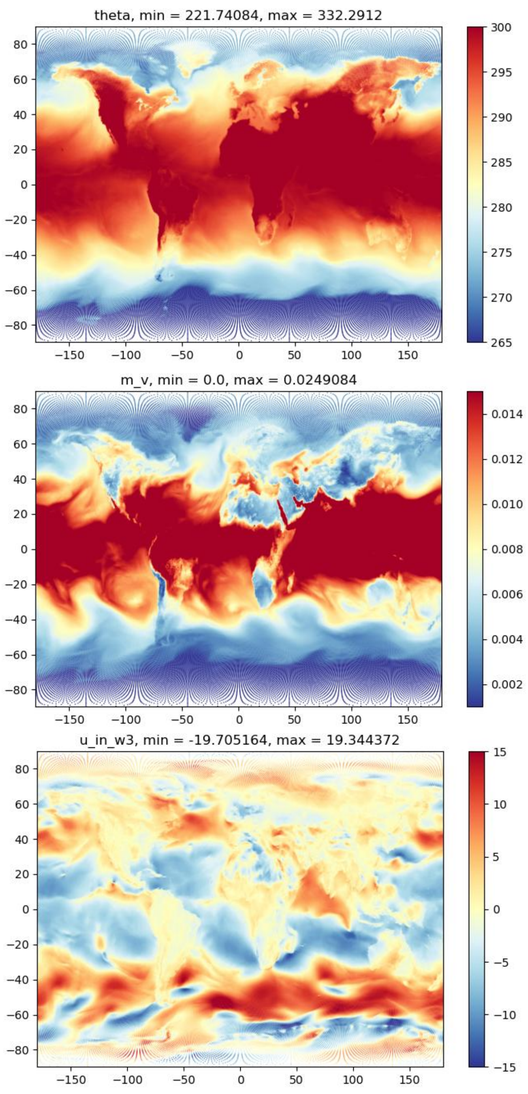
NWP run with AdHImEx in the vertical

- UKMO LFRic (atmosphere-only)[6]
- SWIFT for horizontal transport, finite-element in space, semi-implicit in time
- C224 (~40km grid spacing)
- 70 vertical levels, model top @80km
- Initial conditions: 02-06-2021
- Δt = 12min, T = 24hs
- $C_{max,horiz} \cong \boldsymbol{C_{max,vert}} \cong \mathbf{3}$

Shown are the lowest model levels of

- Potential temperature (theta)
- Moisture (m_v)
- Zonal wind (u_in_w3)

The results are very similar to those obtained with the LFRic flux-form semi-Lagrangian scheme in the vertical (not shown), indicating that it could be suitable to use in atmospheric models.



Conclusion

AdImEx advection schemes combine conservation, stability, and efficiency for large time steps. The new AdHImEx scheme improves their accuracy for large time steps from first- to second-order, using only one matrix solution. It is stable for large C, applicable to arbitrary grids, conserves mass, and preserves constant fields. Its results are accurate and robust even with large C.

Future work will

- Implement AdHImEx in ECMWF PMAP model in the horizontal,
- Add directionality to θ to improve efficiency and accuracy while ensuring stability, and
- Test other high-order implicit time stepping schemes.

References

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Acknowledgements and Contact Information

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